

file as follows:

```
set scada default tcp action block
set scada default udp action block
set scada default icmp action open
set scada maxfds 35
set scada uptime 23456787
add scada tcp port 21 "python plc/honeyd-ftp.py"
add scada tcp port 23 "python plc/honeyd-telnet.py"
add scada tcp port 502 "python plc/honeyd-modbus.py"
add scada tcp port 80 "python plc/honeyd-html.py"
set scada ethernet "08:11:22:33:44:55"
bind 192.168.1.11 scada
```

Table 2 gives an explanation of important configuration options. A typical SCADA system uses FTP, Telnet, HTTP and MODBUS services running on TCP ports 21, 23, 80 and 502 respectively. These are assigned to the corresponding Python scripts in the configuration file.

Table 2

Command	Explanation
bind 192.168.1.11 scada	Used to bind scada honeypot to static IP address 192.168.1.11
add scada tcp port 21 "python plc/honeyd-ftp.py"	Probes towards port 21 (ftp) of the honeypot defined by scada.conf are responded by honeyd-ftp.py script

Similarly, three other Python scripts define responses for port 23 (telnet), port 80 (http) and port 502 (MODBUS)

To start the honeypot configured in scada.conf under daemon mode, use the following command:

```
sudo honeyd -d -f scada.conf
```

Testing the SCADA honeypot

Using nmap for scanning: nmap -n 192.168.1.11 reveals only three open ports: 21, 23 and 80. By default, nmap scans for 1000 well-known ports listed in the nmap-services file. This file does not include port 502 used by the MODBUS protocol.

To scan all TCP ports, use the following command:

```
sudo nmap -p1-65535 -n 192.168.1.11
```

After detecting FTP, Telnet and HTTP ports open; try to use the respective clients to access content from these ports.

Port 80 – the browser: Open the honeypot IP on any Web browser to see the PLC Web page with *Diagnostics*, *Statistics* and *Protocols Supported* menus.

Port 23 Telnet: Telnet to the honeypot IP and establish a connection.

Checking logs on the honeypot

All the traffic received on the scada.conf interface is logged in the `/var/log/scadahoneynet.log` file, which you can study and



Figure 5: PLC home page



Figure 6: Protocols supported

analyse—make sure to allow write permission to this file for the user running honeyd.

The SCADA honeypot project satisfies the basic requirements of a honeypot:

- Appearing as part of a network, though actually isolated
- All access logs are saved for further study
- Its Web interface contains a page that an attacker could perceive to be of great value

Word of caution

Various issues related to the legality of honeypots and honeynets have already been discussed – search the Web for more details. Please make sure to evaluate and understand a particular honeypot by testing it in a lab environment. Do not forget to understand the legal consequences before deploying it in live environments. As an example, if an attacker uses a honeypot to further launch attacks on third party systems, the liability may lie with the honeypot owner. **END** 

References

- [1] Honeyd: <http://www.honeyd.org/>
- [2] SCADA Honeynet project: <http://scadahoneynet.sourceforge.net/>
- [3] Honeypots: are they legal? <http://www.symantec.com/connect/articles/honeypots-are-they-illegal>

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